

CNT sensor experiments (checkpoint)

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Marco Solime, National Institute of Informatics, 10 Feb 2026

Context

Low-Resolution 3D Reconstruction with SAX-NeRF under Limited Detector Geometry

- SAX-NeRF for CT inverse problem
- **Focus**: effect of detector **vertical resolution** on 3D reconstruction
- Controlled experiments on synthetic data

Structure-Aware Sparse-View X-ray 3D Reconstruction

Yuanhao Cai, Jiahao Wang, Alan Yuille^{†,*}, Zongwei Zhou*, Angtian Wang
Johns Hopkins University

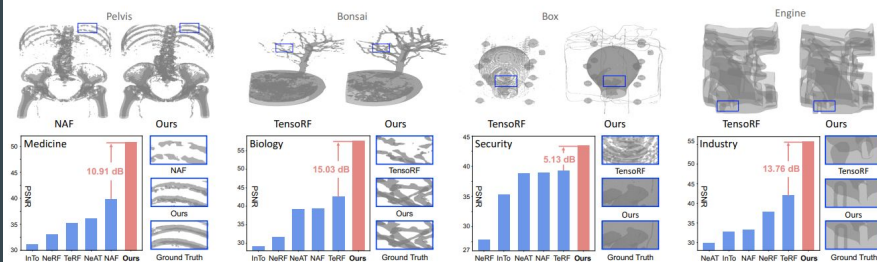
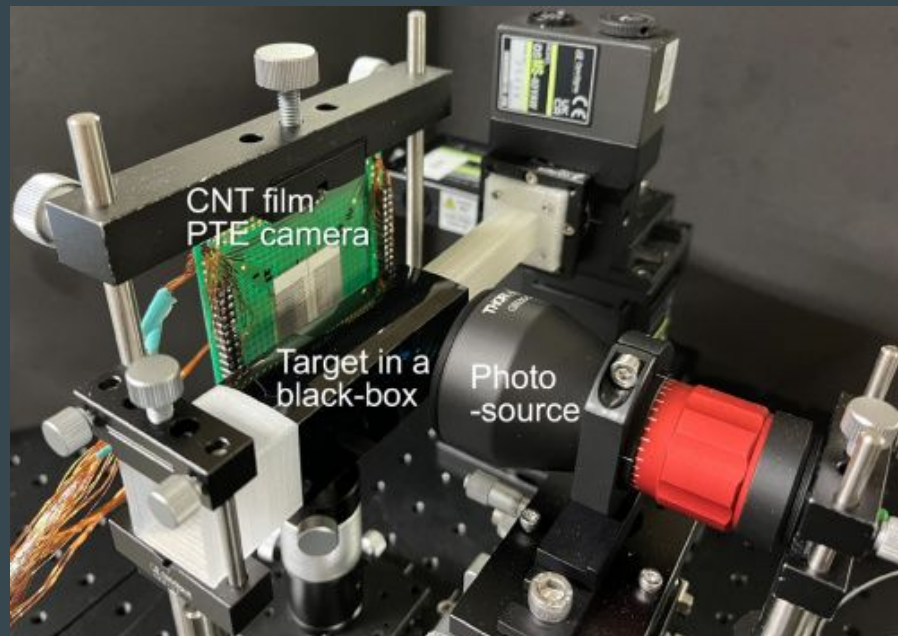


Figure 1. Comparisons of X-ray novel view synthesis. On the collected X3D dataset, our method surpasses state-of-the-art algorithms including InTo (InTomo [56]), NeRF [35], NeAT [43], NAF [59], and TeRF (TensorRF [15]) by **10.91, 15.03, 5.13, and 13.76 dB** in PSNR on the scenes of medicine, biology, security, and industry. The average gains are **over 12 dB**. The visual comparisons of our method and the second-best algorithms on four scenes (pelvis, bonsai, box, and engine) show that our method yields more perceptually pleasing results.

SAX-NeRF, Cai et al, CVPR 2024

What problem are we addressing?

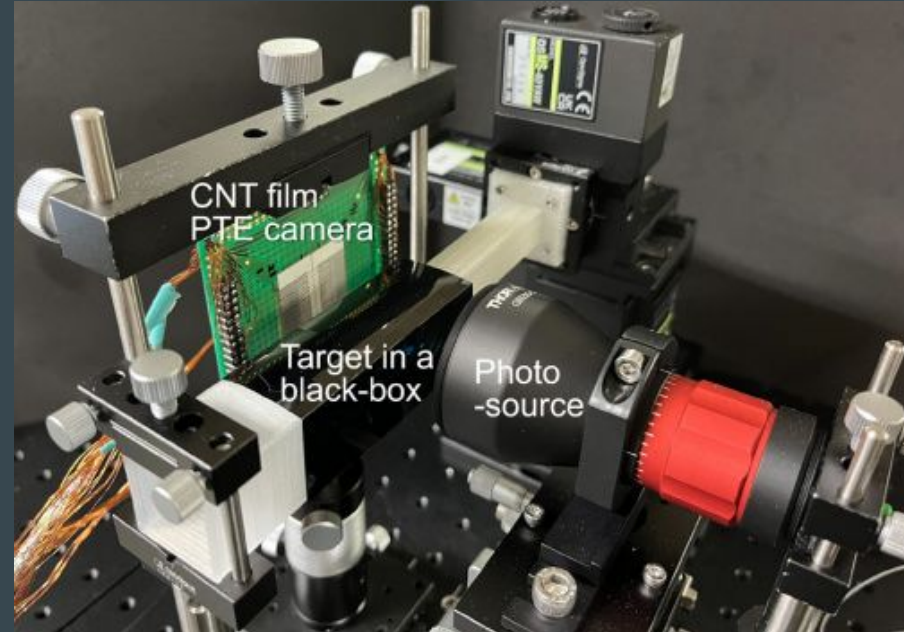
- Reconstruct a **3D attenuation volume** from:
 - Very **low-resolution** projections
 - Extremely **limited detector geometry**
- Motivation:
 - CNT-based sensors with **small vertical pixel count**
 - Sectional / sparse-view CT scenarios
- Key question:
 - *How small can the detector be before SAX-NeRF fails?*



Experimental VH setups, Kubota et al., OPTICA, 2025

Experimental goals

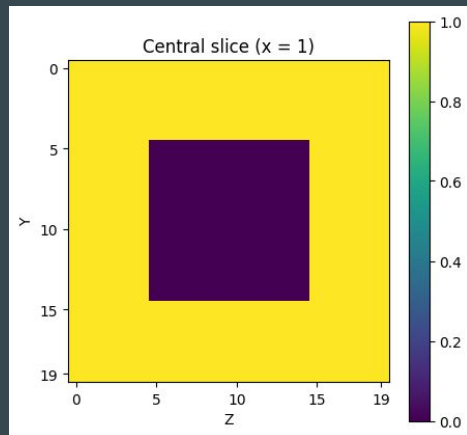
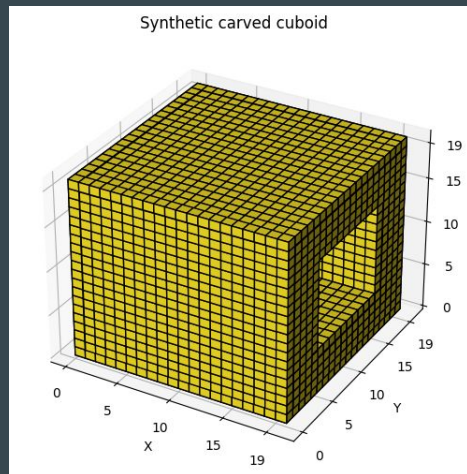
- Quantify reconstruction quality as detector height decreases
- Identify the **minimum detector vertical size** needed for stable 3D recovery
- Separate:
 - Good projection matching
 - vs. meaningful 3D volume reconstruction



Experimental VH setups, Kubota et al., OPTICA, 2025

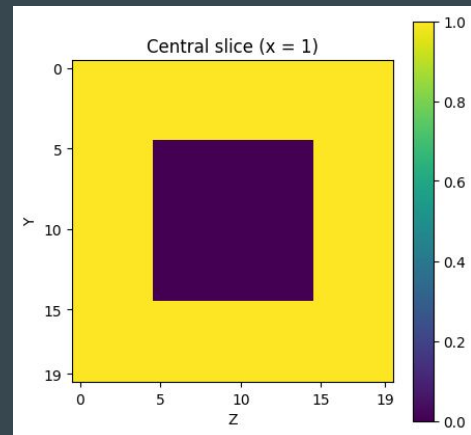
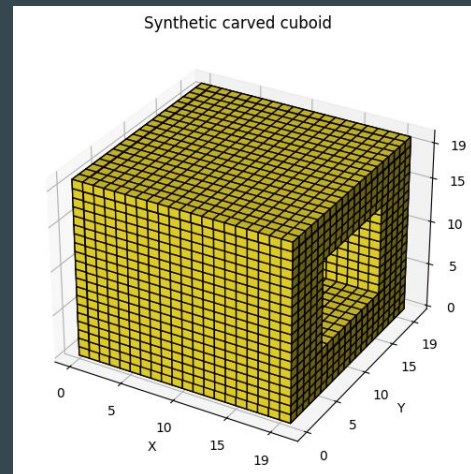
Synthetic carved cuboid (Test Object)

- Volume size: $20 \times 20 \times 20$
- Geometry:
 - Solid cuboid
 - Central void: 10×10 square hole
- Advantages:
 - Simple geometry
 - Sharp edges
 - Easy to diagnose hallucinations and artifacts



CT simulation parameters

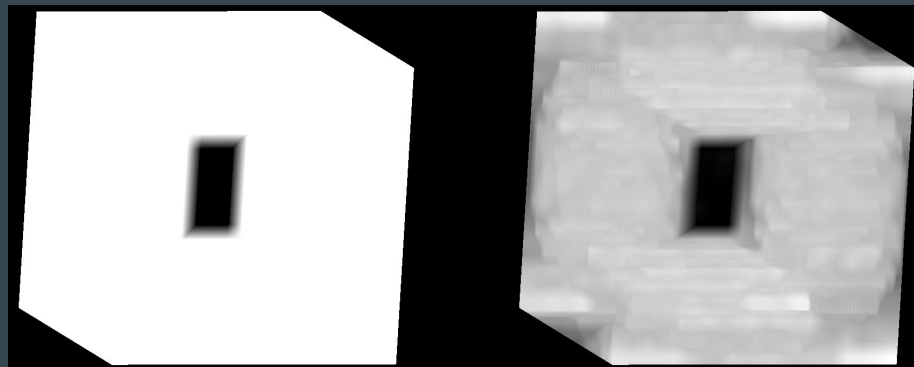
- Acquisition:
 - Parallel beam
 - 360 projections (full rotation)
- Voxel grid:
 - $20 \times 20 \times 20$
- Detector:
 - Horizontal size fixed at 20 pixels
 - Vertical size varied:
 - 20×20
 - 10×20
 - 3×20
- Noise-free projections



Case 1: Detector = 20×20

- Projection predictions:
 - Excellent match to reference
- 3D reconstruction
 - Correct geometry
 - Sharp edges
 - Void clearly resolved

3D volumes



Reference

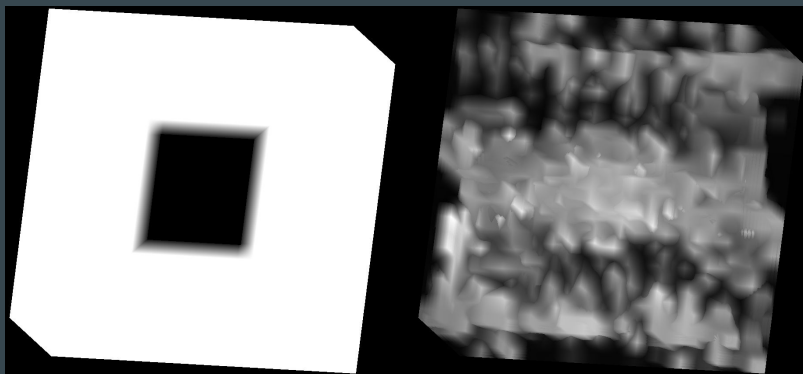
Predicted



Case 2: Detector 10×20

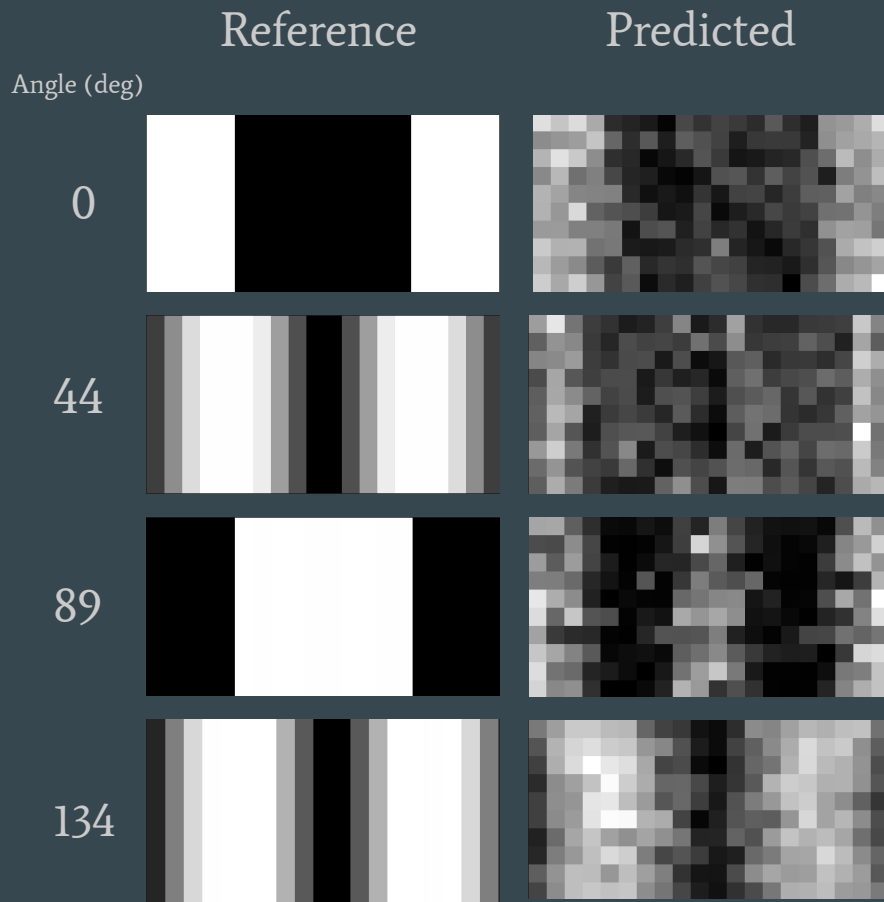
- Projection predictions:
 - Increasing noise
 - Hallucinations
- 3D reconstruction
 - Noisy volume
 - Blurred structure

3D volumes



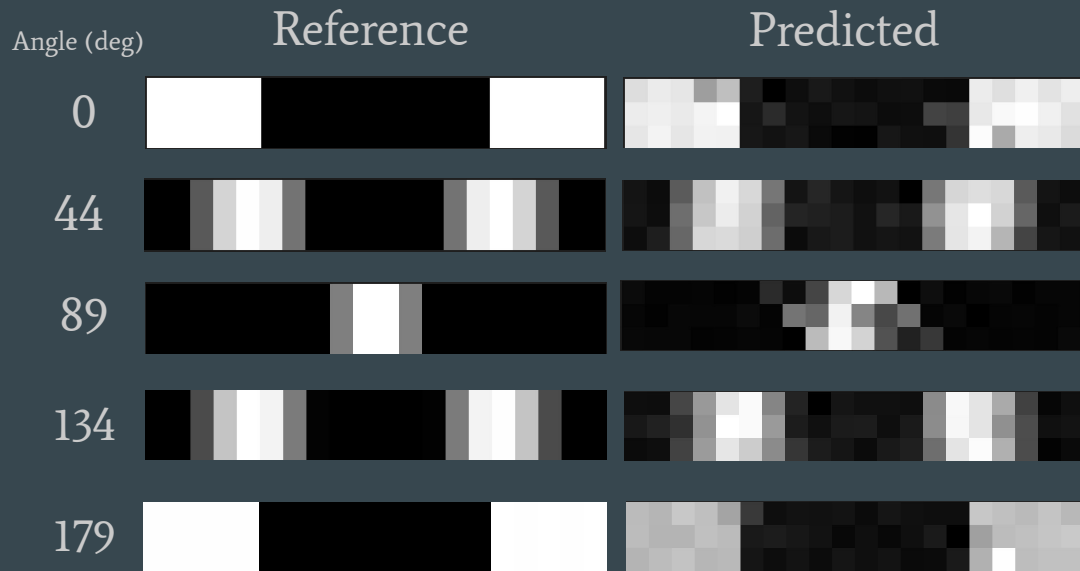
Reference

Predicted

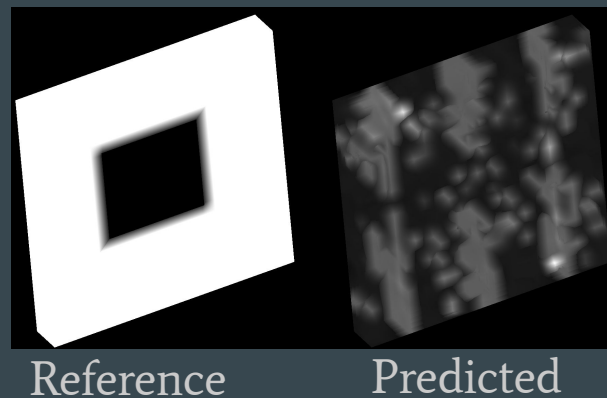


Case 3: Detector 3 × 20

- Projection predictions:
 - Match references surprisingly well
- 3D reconstruction
 - Dominated by noise
 - Loss of shape consistency
 - Strong hallucination artifacts
- Key observation:
 - Good image fit ≠ correct 3D volume
- Failure mode
 - Insufficient vertical information constrain depth



3D volumes



Effect on detector height

Detector size	Projections	3D Volume
20 × 20	✓ Accurate	✓ Correct
10 × 20	✗ Noisy	✗ Hallucinated
3 × 20	✓ Accurate	✗ Hallucinated

- SAX-NeRF optimizes projection consistency first
- Depth ambiguity explodes when vertical rays collapse

Main takeaway

- SAX-NeRF requires a **minimum vertical detector size** (~20 pixels) to successfully reconstruct meaningful 3D structure
- Below this threshold:
 - Network matches projections
 - But **hallucinates density** along depth
- This is a **geometric limitation** , not just a network capacity issue